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$\sqrt{3}$ as a quotient of the two integers $63 - x^2 - 3y^2$ and $36 + 2xy$, which is not possible because $\sqrt{3}$ is an irrational number. To be more explicit, $63 - 36\sqrt{3} = x^2 + 2xy\sqrt{3} + 3y^2$ implies $63 - x^2 - 3y^2 = (36 + 2xy)\sqrt{3}$, and if $36 + 2xy \neq 0$, then we could divide both sides of the equation $63 - x^2 - 3y^2 = (36 + 2xy)\sqrt{3}$ by $36 + 2xy$ to get $\dfrac{63 - x^2 - 3y^2}{36 + 2xy} = \sqrt{3}$.

The correct answer is A.

•

There are 10 books on a shelf, of which 4 are paperbacks and 6 are hardbacks. How many possible selections of 5 books from the shelf contain at least one paperback and at least one hardback?

A. 75

B. 120

C. 210

D. 246

E. 252

Arithmetic **Elementary Combinatorics**

The number of selections of 5 books containing at least one paperback and at least one hardback is equal to $T - N$, where T is the total number of selections of 5 books and N is the number of selections that do not contain both a paperback and a hardback. The value of T is

$$\begin{aligned}\binom{10}{5} &= \frac{10!}{5!\,(10-5)!} = \frac{(6)\,(7)\,(8)\,(9)\,(10)}{(1)\,(2)\,(3)\,(4)\,(5)} \\ &= (7)\,(2)\,(9)\,(2) = 252.\end{aligned}$$

To find the value of N, first note that no selection of 5 books can contain all paperbacks, since there are only 4 paperback books. Thus, the value of N is equal to the number of selections of 5 books that contain all hardbacks, which is equal to 6 since there are 6 ways that a single hardback can be left out when choosing the 5 hardback books. It follows that the number of selections of 5 books containing at least one paperback and at least one hardback is $T - N = 252 - 6 = 246$.

The correct answer is D.

•

If x is to be chosen at random from the set {1, 2, 3, 4} and y is to be chosen at random from the set {5, 6, 7}, what is the probability that xy will be even?

A. $\frac{1}{6}$

B. $\frac{1}{3}$

C. $\frac{1}{2}$

D. $\frac{2}{3}$

E. $\frac{5}{6}$

Arithmetic; Algebra Probability; Concepts of Sets

By the principle of multiplication, since there are 4 elements in the first set and 3 elements in the second set, there are (4)(3) = 12 possible products of xy, where x is chosen from the first set and y is chosen from the second set. These

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(4)(3) = 12 possible products of xy, where x is chosen from the first set and y is chosen from the second set. These products will be even EXCEPT when both x and y are odd. Since there are 2 odd numbers in the first set and 2 odd numbers in the second set, there are (2)(2) = 4 products of x and y that are odd. This means that the remaining 12 − 4 = 8 products are even. Thus, the probability that xy is even is $\frac{18}{12} = \frac{2}{3}$.

The correct answer is D.

•

The function f is defined for each positive three-digit integer n by $f(n) = 2^x\ 3^y\ 5^z$, where x, y, and z are the hundreds, tens, and units digits of n, respectively. If m and v are three-digit positive integers such that $f(m) = 9f(v)$, then $m - v =$

A. 8

B. 9

C. 18

D. 20

E. 80

Algebra Place Value

Let the hundreds, tens, and units digits of m be A, B, and C, respectively; and let the hundreds, tens, and units digits of v be a, b, and c, respectively. From $f(m) = 9f(v)$ it follows that $2^A 3^B 5^C = 9\left(2^a 3^b 5^c\right) = 3^2\left(2^a 3^b 5^c\right) = 2^a 3^{b+2} 5^c$. Therefore, $A = a$, $B = b + 2$, and $C = c$. Now calculate $m - v$.

$m - v$	$= (100A + 10B + C) - (100a + 10b + c)$	place value property
	$= (100a + 10(b + 2) + c) - (100a + 10b + c)$	obtained above
	$= 10(b + 2) - 10b$	combine like terms
	$= 10b + 20 - 10b$	distributive property
	$= 20$	combine like terms

The correct answer is D.

•

If $10^{50} - 74$ is written as an integer in base 10 notation, what is the sum of the digits in that integer?

A. 424

B. 433

C. 440

D. 449

E. 467

Arithmetic Properties of Numbers

$10^2 - 74$	=	100 – 74	=	26
$10^3 - 74$	=	1,000 – 74	=	926
$10^4 - 74$	=	10,000 – 74	=	9,926

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$10^5 - 74$	=	100,000 – 74	=	99,926
$10^6 - 74$	=	1,000,000 – 74	=	999,926

From the table above it is clear that $10^{50} - 74$ in base 10 notation will be 48 digits of 9 followed by the digits 2 and 6. Therefore, the sum of the digits of $10^{50} - 74$ is equal to 48(9) + 2 + 6 = 440.

The correct answer is C.

•

A certain company that sells only cars and trucks reported that revenues from car sales in 1997 were down 11 percent from 1996 and revenues from truck sales in 1997 were up 7 percent from 1996. If total revenues from car sales and truck sales in 1997 were up 1 percent from 1996, what is the ratio of revenue from car sales in 1996 to revenue from truck sales in 1996?

A. 1:2

B. 4:5

C. 1:1

D. 3:2

E. 5:3

Algebra; Arithmetic First-Degree Equations; Percents

Let C_{96} and C_{97} represent revenues from car sales in 1996 and 1997, respectively, and let T_{96} and T_{97} represent revenues from truck sales in 1996 and 1997, respectively. A decrease of 11 percent in revenue from car sales from 1996 to 1997 can be represented as $(1 - 0.11)C_{96} = C_{97}$, and a 7 percent increase in revenue from truck sales from 1996 to 1997 can be represented as $(1 + 0.07)T_{96} = T_{97}$.

An overall increase of 1 percent in revenue from car and truck sales from 1996 to 1997 can be represented as $C_{97} + T_{97} = (1 + 0.01)(C_{96} + T_{96})$.

Then, by substitution of expressions for C_{97} and T_{97} that were derived above, $(1 - 0.11)C_{96} + (1 + 0.07)T_{96} = (1 + 0.01)(C_{96} + T_{96})$ and so $0.89C_{96} + 1.07T_{96} = 1.01(C_{96} + T_{96})$ or $0.89C_{96} + 1.07T_{96} = 1.01C_{96} + 1.01T_{96}$.

Then, combining like terms gives $(1.07 - 1.01)T_{96} = (1.01 - 0.89)C_{96}$ or $0.06T_{96} = 0.12C_{96}$. Thus $\frac{C_{96}}{T_{96}} = \frac{0.06}{0.12} = \frac{1}{2}$. The ratio of revenue from car sales in 1996 to revenue from truck sales in 1996 is 1:2.

The correct answer is A.

•

Becky rented a power tool from a rental shop. The rent for the tool was $12 for the first hour and $3 for each additional hour. If Becky paid a total of $27, excluding sales tax, to rent the tool, for how many hours did she rent it?

A. 5

B. 6

C. 9

D. 10

E. 12

Arithmetic Applied Problems

Becky paid a total of $27 to rent the power tool. She paid $12 to rent the tool for the first hour and $27 – $12 = $15 to rent the tool for the additional hours at the rate of $3 per additional hour. It follows that she rented the tool for $\frac{15}{3} = 5$ additional hours and a total of 1 + 5 = 6 hours.

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The correct answer is B.

•

If $4 < \frac{7-x}{3}$ which of the following must be true?

I. $5 < x$

II. $|x+3| > 2$

III. $-(x+5)$ is positive.

A. II only

B. III only

C. I and II only

D. II and III only

E. I, II, and III

Algebra Inequalities

Given that $4 < \frac{7-x}{3}$, it follows that $12 < 7 - x$. Then $5 < -x$ or, equivalently, $x < -5$.

I. If $4 < \frac{7-x}{3}$, then $x < -5$. If $5 < x$ were true, then, by combining $5 < x$ and $x < -5$, it would follow that $5 < -5$, which cannot be true. Therefore, it is not the case that, if $4 < \frac{7-x}{3}$, then Statement I must be true. In fact, Statement I is never true.

II. If $4 < \frac{7-x}{3}$, then $x < -5$, and it follows that $x + 3 < -2$. Since $-2 < 0$, then $x + 3 < 0$ and $|x+3| = -(x+3)$. If $x + 3 < -2$, then $-(x+3) > 2$ and by substitution, $|x+3| > 2$. Therefore, Statement II must be true for every value of x such that $x < -5$. Therefore, Statement II must be true if $4 < \frac{7-x}{3}$.

III. If $4 < \frac{7-x}{3}$, then $x < -5$ and $x + 5 < 0$. But if $x + 5 < 0$, then it follows that $-(x+5) > 0$ and so $-(x+5)$ is positive. Therefore Statement III must be true if $4 < \frac{7-x}{3}$.

The correct answer is D.

•

On a certain day, a bakery produced a batch of rolls at a total production cost of \$300. On that day, $\frac{4}{5}$ of the rolls in the batch were sold, each at a price that was 50 percent greater than the average (arithmetic mean) production cost per roll. The remaining rolls in the batch were sold the next day, each at a price that was 20 percent less than the price of the day before. What was the bakery's profit on this batch of rolls?

A. \$150

B. \$144

C. \$132

D. \$108

E. \$90

Arithmetic Applied Problems

Let n be the number of rolls in the batch and p be the average production price, in dollars, per roll. Then the total cost of the batch is $np = 300$ dollars, and the total revenue from selling the rolls in the batch is

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of the batch is $np = 300$ dollars, and the total revenue from selling the rolls in the batch is

$$\left(\frac{4}{5}n\right)(1.5p) + \left(\frac{1}{5}n\right)(0.8)(1.5p) = \left(\frac{4}{5}n\right)\left(\frac{3}{2}p\right) + \left(\frac{1}{5}n\right)\left(\frac{4}{5}\right)\left(\frac{3}{2}p\right) = \left(\frac{6}{5} + \frac{6}{25}\right)np = \left(\frac{36}{25}\right)np$$

. Therefore, the profit from selling the rolls in the batch is $\left(\frac{36}{25}\right)np - np = \left(\frac{11}{25}\right)np = \left(\frac{11}{25}\right)(300)$ dollars = 132 dollars.

The correct answer is C.

•

A set of numbers has the property that for any number t in the set, $t + 2$ is in the set. If –1 is in the set, which of the following must also be in the set?

I. –3

II. 1

III. 5

A. I only

B. II only

C. I and II only

D. II and III only

E. I, II, and III

Arithmetic Properties of Numbers

It is given that –1 is in the set and, if t is in the set, then $t + 2$ is in the set.

I. Since {–1, 1, 3, 5, 7, 9, 11, …} contains –1 and satisfies the property that if t is in the set, then $t + 2$ is in the set, it is not true that –3 must be in the set.

II. Since –1 is in the set, −1 + 2 = 1 is in the set. Therefore, it must be true that 1 is in the set.

III. Since –1 is in the set, −1 + 2 = 1 is in the set. Since 1 is in the set, 1 + 2 = 3 is in the set. Since 3 is in the set, 3 + 2 = 5 is in the set. Therefore, it must be true that 5 is in the set.

The correct answer is D.

•

A couple decides to have 4 children. If they succeed in having 4 children and each child is equally likely to be a boy or a girl, what is the probability that they will have exactly 2 girls and 2 boys?

A. $\frac{3}{8}$

B. $\frac{1}{4}$

C. $\frac{3}{16}$

D. $\frac{1}{18}$

E. $\frac{1}{16}$

Arithmetic Probability

Representing the birth order of the 4 children as a sequence of 4 letters, each of which is B for boy and G for girl, there are 2 possibilities (B or G) for the first letter, 2 for the second letter, 2 for the third letter, and 2 for the fourth letter, making a total of $2^4 = 16$ sequences. The table below categorizes some of these 16 sequences.

# of	# of		# of

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# of boys	# of girls	Sequences	# of sequences
0	4	GGGG	1
1	3	BGGG, GBGG, GGBG, GGGB	4
3	1	GBBB, BGBB, BBGB, BBBG	4
4	0	BBBB	1

The table accounts for 1 + 4 + 4 + 1 = 10 sequences. The other 6 sequences will have 2Bs and 2Gs. Therefore the probability that the couple will have exactly 2 boys and 2 girls is $\frac{6}{16} = \frac{3}{8}$. For the mathematically inclined, if it is assumed that a couple has a fixed number of children, that the probability of having a girl each time is p, and that the sex of each child is independent of the sex of the other children, then the number of girls, x, born to a couple with n children is a random variable having the binomial probability distribution. The probability of having exactly x girls born to a couple with n children is given by the formula $\binom{n}{x} p^x (1-p)^{n-x}$. For the problem at hand, it is given that each child is equally likely to be a boy or a girl, and so $p = \frac{1}{2}$. Thus, the probability of having exactly 2 girls born to a couple with 4 children is

$$\binom{4}{2}\left(\frac{1}{2}\right)^2\left(\frac{1}{2}\right)^2 = \frac{4!}{2!2!}\left(\frac{1}{2}\right)^2\left(\frac{1}{2}\right)^2 = (6)\left(\frac{1}{4}\right)\left(\frac{1}{4}\right) = \frac{6}{16} = \frac{3}{8}.$$

The correct answer is A.

•

The closing price of Stock X changed on each trading day last month. The percent change in the closing price of Stock X from the first trading day last month to each of the other trading days last month was less than 50 percent. If the closing price on the second trading day last month was $10.00, which of the following CANNOT be the closing price on the last trading day last month?

A. $3.00

B. $9.00

C. $19.00

D. $24.00

E. $29.00

Arithmetic Applied Problems; Percents

Let P be the first-day closing price, in dollars, of the stock. It is given that the second-day closing price was $(1+n\%)P = 10$, so $P = \frac{10}{1+n\%}$, for some value of n such that $-50 < n < 50$. Therefore, P is between $\frac{10}{1+0.50} \approx 6.67$ and $\frac{10}{1-0.50} = 20$. Hence, if Q is the closing price, in dollars, of the stock on the last day, then Q is between (0.50)(6.67) ≈ 3.34 (50% decrease from the lowest possible first-day closing price) and (1.50)(20) = 30 (50% increase from the greatest possible first-day closing price). The only answer choice that gives a number of dollars not between 3.34 and 30 is the first answer choice.

The correct answer is A.

•

How many different positive integers are factors of 441?

A. 4

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B. 6

C. 7

D. 9

E. 11

Arithmetic Properties of Numbers

Like any integer, 441 has itself and 1 as factors. To find all its other positive factors, we first use prime factorization to write 441 as a product of its prime factors: 441 = 3 × 3 × 7 × 7. Then we can group those two 3s and two 7s in various ways to find all their products, which will be the other positive factors of 441. That is, 441 = 3 × (3 × 7 × 7) = (3 × 3) × (7 × 7) = (3 × 7) × (3 × 7) = (3 × 3 × 7) × 7. This shows that 441 has 7 different positive integer factors other than itself and 1. These are 3, 7, 3 × 3 = 9, 3 × 7 = 21, 7 × 7 = 49, 3 × 3 × 7 = 63, and 3 × 7 × 7 = 147. So altogether, 441 has 9 different positive integers as factors: 1, 3, 7, 9, 21, 49, 63, 147, and 441.

The correct answer is D.

•

If n is a positive integer and n^2 is divisible by 72, then the largest positive integer that must divide n is

A. 6

B. 12

C. 24

D. 36

E. 48

Arithmetic Properties of Numbers

Since n^2 is divisible by 72, $n^2 = 72k$ for some positive integer k. Since $n^2 = 72k$, then $72k$ must be a perfect square. Since $72k = (2^3)(3^2)k$, then $k = 2m^2$ for some positive integer m in order for $72k$ to be a perfect square. Then, $n^2 = 72k = (2^3)(3^2)(2m^2) = (2^4)(3^2)m^2 = \left[(2^2)(3)(m)\right]^2$, and $n = (2^2)(3)(m)$. The positive integers that MUST divide n are 1, 2, 3, 4, 6, and 12. Therefore, the largest positive integer that must divide n is 12.

The correct answer is B.

•

A certain grocery purchased x pounds of produce for p dollars per pound. If y pounds of the produce had to be discarded due to spoilage and the grocery sold the rest for s dollars per pound, which of the following represents the gross profit on the sale of the produce?

A. $(x - y)s - xp$

B. $(x - y)p - ys$

C. $(s - p)y - xp$

D. $xp - ys$

E. $(x - y)(s - p)$

Algebra Simplifying Algebraic Expressions; Applied Problems

Since the grocery bought x pounds of produce for p dollars per pound, the total cost of the produce was xp dollars. Since y pounds of the produce was discarded, the grocery sold $x - y$ pounds of produce at the price of s dollars per pound, yielding a total revenue of $(x - y)s$ dollars. Then, the grocery's gross profit on the sale of the produce is its total revenue minus its total cost or $(x - y)s - xp$ dollars.

The correct answer is A.

•

If x, y, and z are positive integers such that x is a factor of y, and x is a multiple of z, which of the following is NOT necessarily an integer?

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A. $\frac{x+z}{z}$

B. $\frac{y+z}{x}$

C. $\frac{x+y}{z}$

D. $\frac{xy}{z}$

E. $\frac{yz}{x}$

Arithmetic Properties of Numbers

Since the positive integer x is a factor of y, then $y = kx$ for some positive integer k. Since x is a multiple of the positive integer z, then $x = mz$ for some positive integer m.

Substitute these expressions for x and/or y into each answer choice to find the one expression that is NOT necessarily an integer.

A. $\frac{x+z}{z} = \frac{mz+z}{z} = \frac{(m+1)z}{z} = m+1$, which MUST be an integer

B. $\frac{y+z}{x} = \frac{y}{x} + \frac{z}{x} = \frac{kx}{x} + \frac{z}{mz} = k + \frac{1}{m}$, which NEED NOT be an integer

Because only one of the five expressions need not be an integer, the expressions given in C, D, and E need not be tested. However, for completeness,

C. $\frac{x+y}{z} = \frac{mz+kx}{z} = \frac{mz+k(mz)}{z} = \frac{mz(1+k)}{z} = m(1+k)$, which MUST be an integer

D. $\frac{xy}{z} = \frac{(mz)y}{z} = my$, which MUST be an integer

E. $\frac{yz}{x} = \frac{(kx)(z)}{x} = kz$, which MUST be an integer

The correct answer is B.

•

Running at their respective constant rates, Machine X takes 2 days longer to produce w widgets than Machine Y. At these rates, if the two machines together produce $\frac{5}{4}w$ widgets in 3 days, how many days would it take Machine X alone to produce $2w$ widgets?

A. 4

B. 6

C. 8

D. 10

E. 12

Algebra Applied Problems

If x, where $x > 2$, represents the number of days Machine X takes to produce w widgets, then Machine Y takes $x-2$ days to produce w widgets. It follows that Machines X and Y can produce $\frac{w}{x}$ and $\frac{w}{x-2}$ widgets, respectively, in 1 day and together they can produce $\frac{w}{x} + \frac{w}{x-2}$ widgets in 1 day. Since it is given that, together, they can produce $\frac{5}{4}w$ widgets in 3 days, it follows that, together, they can produce $\frac{1}{3}\left(\frac{5}{4}w\right) = \frac{5}{12}w$ widgets in 1

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they can produce $\frac{1}{4}w$ widgets in 3 days, it follows that, together, they can produce $\frac{1}{3}\left(\frac{1}{4}w\right) = \frac{1}{12}w$ widgets in 1 day. Thus,

$$\begin{aligned}
\frac{w}{x} + \frac{w}{x-2} &= \frac{5}{12}w \\
\left(\frac{1}{x} + \frac{1}{x-2}\right)w &= \frac{5}{12}w \\
\left(\frac{1}{x} + \frac{1}{x-2}\right) &= \frac{5}{12} \\
12x(x-2)\left(\frac{1}{x} + \frac{1}{x-2}\right) &= 12x(x-2)\left(\frac{5}{12}\right) \\
12[(x-2)+x] &= 5x(x-2) \\
12(2x-2) &= 5x(x-2) \\
24x - 24 &= 5x^2 - 10x \\
0 &= 5x^2 - 34x + 24 \\
0 &= (5x-4)(x-6) \\
x &= \frac{4}{5} \text{ or } 6
\end{aligned}$$

Therefore, since $x > 2$, it follows that $x = 6$. Machine X takes 6 days to produce w widgets and 2(6) = 12 days to produce $2w$ widgets.

The correct answer is E.

•

What is the greatest positive integer n such that 5^n divides $10! - (2)(5!)^2$?

A. 2

B. 3

C. 4

D. 5

E. 6

Arithmetic Properties of Numbers; Exponents

The greatest positive integer n such that 5^n divides a given integer is the number of factors of 5 in the prime factorization of the given integer. By repeated identification of common factors, the indicated difference can be factored sufficiently to determine the number of factors of 5 in its prime factorization. In the computations that follow, we have used the equalities 10! = (5!)(6)(7)(8)(9)(10) and 5! = (2)(3)(4)(5).

$10! - (2)(5!)^2$

$= (5! \cdot 6 \cdot 7 \cdot 8 \cdot 9 \cdot 10) - (2 \cdot 5! \cdot 5!)$

$= (5!)(6 \cdot 7 \cdot 8 \cdot 9 \cdot 10 - 2 \cdot 5!)$

$= (5!)(6 \cdot 7 \cdot 8 \cdot 9 \cdot 10 - 2 \cdot 2 \cdot 3 \cdot 4 \cdot 5)$

$= (5!)(2^4)(3 \cdot 7 \cdot 9 \cdot 10 - 3 \cdot 5)$

$= (5!)(2^4)(3)(7 \cdot 9 \cdot 10 - 5)$

$= (5!)(2^4)(3)(5)(7 \cdot 9 \cdot 2 - 1)$

$= (5!)(2^4)(3)(5)(63 \cdot 2 - 1)$

$= (5!)(2^4)(3)(5)(126 - 1)$

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= (5!)(2^4)(3)(5)(125)

Since there is exactly 1 factor of 5 in 5! = (2)(3)(4)(5), no factors of 5 in either 2^4 or 3, exactly 1 factor of 5 in 5, and exactly 3 factors of 5 in 125 = 5^3, it follows that there are 1 + 1 + 3 = 5 factors of 5 in 10! – (2)(5!)^2 = (5!)(2^4)(3)(5)(125).

The correct answer is D.

•

Yesterday, Candice and Sabrina trained for a bicycle race by riding around an oval track. They both began riding at the same time from the track's starting point. However, Candice rode at a faster pace than Sabrina, completing each lap around the track in 42 seconds, while Sabrina completed each lap around the track in 46 seconds. How many laps around the track had Candice completed the next time that Candice and Sabrina were together at the starting point?

A. 21

B. 23

C. 42

D. 46

E. 483

Arithmetic Applied Problems; Properties of Integers

Let C and S be the number of laps around the track, respectively, whenever Candice and Sabrina were together again at the starting point. Since Candice completes each lap in 42 seconds, Candice had been riding for a total of $42C$ seconds, and since Sabrina completes each lap in 46 seconds, Sabrina had been riding for a total of $46S$ seconds. Because they had been riding for the same total amount of time, we have $42C = 46S$, or $21C = 23S$, where C and S are positive integers.

Because 23 is a prime number that divides the product of 21 and C (note that 23 divides $23S$ and $23S = 21C$), it follows that 23 divides 21 (not true) or 23 divides C, and hence 23 divides C. Also, because 3 is a prime number that divides the product of 23 and S (note that 3 divides $21C$ and $21C = 23S$), it follows that 3 divides 23 (not true) or 3 divides S, and hence 3 divides S. Finally, because 7 is a prime number that divides the product of 23 and S (note that 7 divides $21C$ and $21C = 23S$), it follows that 7 divides 23 (not true) or 7 divides S, and hence 7 divides S

It follows that C is a multiple of 23 and S is a multiple of both 3 and 7. The least positive integer values for C and S with this property are $C = 23$ and $S = 3 \times 7 = 21$. Therefore, the next time after beginning that Candice and Sabrina were together at the starting point, Candice had completed 23 laps and Sabrina had completed 21 laps.

The correct answer is B.

•

If $n = 9! - 6^4$, which of the following is the greatest integer k such that 3^k is a factor of n?

A. 1

B. 3

C. 4

D. 6

E. 8

Arithmetic Properties of Integers

The following charts isolate and count the occurrences of 2 and of 3 in the factorizations of 9! = (9)(8)(7)(6)(5)(4)(3)(2) and 6^4 = (6)(6)(6)(6).

9!	=	(9)	(8)	(7)	(6)	(5)	(4)	(3)	(2)
Occurrences of 2		0	3	0	1	0	2	0	1
Occurrences of 3		2	0	0	1	0	0	1	0

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Occurrences of 3		2	0	0	1	0	0	1	0

6^4	=	(6)	(6)	(6)	(6)
Occurrences of 2		1	1	1	1
Occurrences of 3		1	1	1	1

So, 9! – 6^4 = ($2^7 \cdot 3^4 \cdot 5 \cdot 7$) – ($2^4 \cdot 3^4$) = ($2^4 \cdot 3^4$)($2^3 \cdot 5 \cdot 7 - 1$) = ($2^4 \cdot 3^4$)(279) = ($2^4 \cdot 3^4$)($3^2 \cdot 31$) = $2^4 \cdot 3^6 \cdot 31$, where 31 is prime. Therefore $k = 6$.

Alternatively, express $n = 9! - 6^4$ as (9)(8)(7)(6)(5)(4)(3)(2)(1) – 6^4 = [(9)(8)](7)(6)(5)(4)[(3)(2)](1) – 6^4. Then factor (9)(8) as (36)(2) = (6^2)(2) and multiply (3)(2) to get 6. Factoring 6^4 from $n = (6^2)(2)(7)(6)(5)(4)(6)(1) - 6^4$ gives $n = 6^4\left[(2)(7)(5)(4)(1) - 1\right] = 6^4(279)$. It follows that 6^4 has 4 factors of 3 and 279 has 2 additional factors of 3 since 279 = (3^2)(31), so the greatest integer k such that 3^k is a factor of n is 4 + 2 = 6.

The correct answer is D.

•

The integer 120 has many factorizations. For example, 120 = (2)(60), 120 = (3)(4)(10), and 120 = (–1)(–3)(4)(10). In how many of the factorizations of 120 are the factors consecutive integers in ascending order?

A. 2

B. 3

C. 4

D. 5

E. 6

Arithmetic Properties of Integers

All of the positive factors of 120 listed in ascending order are 1, 2, 3, 4, 5, 6, 8, 10, 12, 15, 20, 24, 30, 40, 60, and 120. The negative factors of 120 listed in ascending order are –120, –60, –40, –30, –24, –20, –15, –12, –10, –8, –6, –5, –4, –3, –2, and –1. Examining these lists for groups of consecutive factors whose product is 120 gives (1)(2)(3)(4)(5), (2)(3)(4)(5), (4)(5)(6), and (–5)(–4)(–3)(–2).

The correct answer is C.

•

Jorge's bank statement showed a balance that was \$0.54 greater than what his records showed. He discovered that he had written a check for $\$x.yz$ and had recorded it as $\$x.zy$, where each of x, y, and z represents a digit from 0 though 9. Which of the following could be the value of z?

A. 2

B. 3

C. 4

D. 5

E. 6

Arithmetic Place Value

Since the amount Jorge recorded for the check ($\$x.zy$) was \$0.54 more than the actual amount of the check ($\$x.yz$), it follows that $\$x.zy - \$x.yz = \$0.54$. This is equivalent to $x + \frac{z}{10} + \frac{y}{100} - \left(x + \frac{y}{10} + \frac{z}{100}\right) = \frac{54}{100}$.

Then $\frac{10z + y}{100} - \frac{10y + z}{100} = \frac{54}{100}$ or, equivalently, $10z + y - (10y + z) = 54$. It follows that $9z - 9y = 54$ or

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100 100 100

$z - y = 6$. Since y and z are digits, the possible values of y and z, respectively, are 0 and 6, 1 and 7, 2 and 8, and 3 and 9. Of the possible values of z, only 6 is given as one of the answer choices.

The correct answer is E.

•

One side of a parking stall is defined by a straight stripe that consists of n painted sections of equal length with an unpainted section $\frac{1}{2}$ as long between each pair of consecutive painted sections. The total length of the stripe from the beginning of the first painted section to the end of the last painted section is 203 inches. If n is an integer and the length, in inches, of each unpainted section is an integer greater than 2, what is the value of n?

A. 5

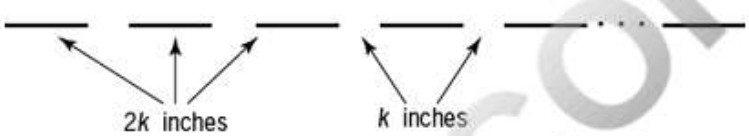
B. 9

C. 10

D. 14

E. 29

Algebra Applied Problems



The figure above is a schematic diagram of the parking stall's painted sections, where each painted section has length $2k$ inches and each unpainted section has length k inches. Since there is a total of n painted sections and a total of $(n-1)$ unpainted sections, it follows that $n(2k) + (n-1)k = 203$, or $(3n-1)k = 203$. Also, since n and k are positive integers with $n \geq 2$, and 203 = (7)(29) is the only factorization of 203 with integer factors greater than or equal to 2, we have two cases.

Case 1: $3n - 1 = 7$ and $k = 29$. In this case we have $n = \frac{8}{3}$.

Case 2: $3n - 1 = 29$ and $k = 7$. In this case we have $n = 10$.

Because $\frac{8}{3}$ is not an integer, we discard Case 1, and hence $n = 10$.

The correct answer is C.

•

This year, Henry will save a certain amount of his income, and he will spend the rest. Next year, Henry will have no income, but for each dollar that he saves this year, he will have $1 + r$ dollars available to spend. In terms of r, what fraction of his income should Henry save this year so that next year the amount he has available to spend will be equal to half the amount that he spends this year?

A. $\frac{1}{r} + 2$

B. $\frac{1}{2r} + 2$

C. $\frac{1}{3r} + 2$

D. $\frac{1}{r} + 3$

E. $\frac{1}{2r} + 3$

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Algebra Sets

Let x be the number of dollars Henry should save this year and y be his total income in dollars this year. Then for him to have an amount available to spend next year equal to half the amount he spends this year, $x(1+r)$ must equal $\frac{y-x}{2}$.

Thus, $(2+2r)x$ must equal $y-x$, so y must equal $(3+2r)x$. Therefore, the fraction of his income he must save this year is $\frac{x}{y} = \frac{x}{(3+2r)x} = \frac{1}{2r+3}$.

The correct answer is E.

Machine	Consecutive Minutes Machine Is Off	Units of Power When On
A	17	15
B	14	18
C	11	12

•

At a certain factory, each of Machines A, B, and C is periodically on for exactly 1 minute and periodically off for a fixed number of consecutive minutes. The table above shows that Machine A is on and uses 15 units of power every 18th minute, Machine B is on and uses 18 units of power every 15th minute, and Machine C is on and uses 12 units of power every 12th minute. The factory has a backup generator that operates only when the total power usage of the 3 machines exceeds 30 units of power. What is the time interval, in minutes, between consecutive times the backup generator begins to operate?

A. 36

B. 63

C. 90

D. 180

E. 270

Arithmetic Applied Problems

The given table shows that the backup generator will not operate when only one of the machines is operating, since none of the machines uses more than 30 units of power. The table below shows the power usage when more than one machine is operating at the same time.

Machines	Units of power when on	Backup generator
A & B	15 + 18 = 33	On
B & C	18 + 12 = 30	Off
C & A	12 + 15 = 27	Off
A & B & C	15 + 18 + 12 = 45	On

Thus, the backup generator will be on whenever Machines A and B are both on, this being true regardless of whether Machine C is on. We are given that Machine A is on for 1 minute every 18 minutes and Machine B is on for 1 minute

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Machine B is on. We are given that Machine A is on for 1 minute every 18 minutes and Machine B is on for 1 minute every 15 minutes. Therefore, if Machines A and B are both on for a certain minute, then the following are the minutes when these machines are again on.

Minutes when Machine A is on: 1st, 19th, 37th, 55th, 73rd, $\mathbf{91^{st}}$, 109th, …

Minutes when Machine B is on: 1st, 16th, 31st, 46th, 61st, 76th, $\mathbf{91^{st}}$, 106th, …

Therefore, the next time Machines A and B are both on is the 91st minute, which is 90 minutes after the first minute.

Alternatively, Machine A is on every $18 = (2)(3^2)$ minutes and Machine B is on every $15 = (3)(5)$ minutes, so the machines are both on every $(2)(3^2)(5) = 90$ minutes (least common multiple of 18 and 15).

The correct answer is C.

•

In a certain region, the number of children who have been vaccinated against rubella is twice the number who have been vaccinated against mumps. The number who have been vaccinated against both is twice the number who have been vaccinated against mumps. If 5,000 have been vaccinated against both, how many have been vaccinated against rubella only?

A. 2,500

B. 7,500

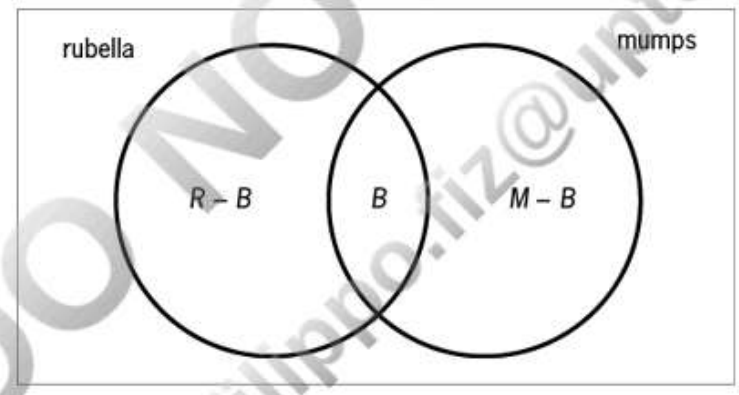
C. 10,000

D. 15,000

E. 17,500

Algebra **Simultaneous Equations; Sets (Venn Diagrams)**

Let R, M, and B be the number, respectively, of children who have been vaccinated against rubella, against mumps, and against both. Then the number of children who have been vaccinated against rubella only is $R - B$ and the number of children who have been vaccinated against mumps only is $M - B$, as shown in the Venn diagram below.



From the given information, we have (i) $R = 2M$, (ii) $B = 2(M - B)$, and (iii) $B = 5{,}000$. Substituting the value of B given in (iii) into (ii) gives $5{,}000 = 2(M - 5{,}000)$, or $M = 7{,}500$. Using this value of M in (i) gives $R = 2(7{,}500) = 15{,}000$. Therefore, the number vaccinated against rubella only is $R - B = 15{,}000 - 5{,}000 = 10{,}000$.

The correct answer is C.

•

Three boxes of supplies have an average (arithmetic mean) weight of 7 kilograms and a median weight of 9 kilograms. What is the maximum possible weight, in kilograms, of the lightest box?

A. 1

B. 2

C. 3

D. 4

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